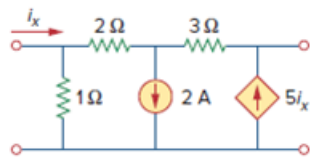


Problem

Obtain the g parameters for the network in Fig. 19.128 using *PSpice* or *MultiSim*.

Figure 19.128



Step-by-step solution

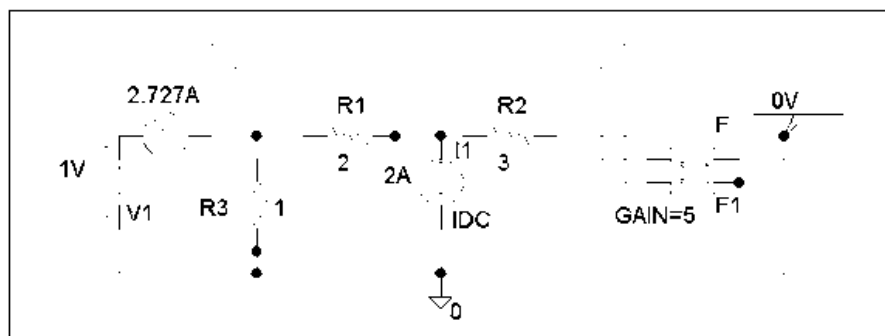
Step 1 of 2

(a) By definition, $g_{11} = \frac{I_1}{V_1} \bigg|_{I_2=0}$, $g_{21} = \frac{V_2}{V_1} \bigg|_{I_2=0}$.

We let $V_1 = 1$ V and open-circuit the output port. The schematic is shown below. After simulation, we obtain

$$g_{11} = I_1 = 2.7$$

$$g_{21} = V_2 = 0.0$$



Step 2 of 2

(b) Similarly,

$$g_{12} = \left. \frac{I_1}{I_2} \right|_{V_1=0}, \quad g_{22} = \left. \frac{V_2}{I_2} \right|_{V_1=0}$$

We let $I_2 = 1$ A and short-circuit the input port. The schematic is shown below. After simulation,

$$g_{12} = I_1 = 0$$

$$g_{22} = V_2 = 0$$

Thus $[g] = \begin{bmatrix} 2.7278 & 0 \\ 0 & 0 \end{bmatrix}$

